

MATH 351 first midterm checklist

General

- I have read all the assigned sections of the textbook.
- I have read all the feedback on Blackboard for my homework assignments.
- I have studied all the solution sets for homework assignments on Blackboard.
- I have studied all the solution sets for quizzes on Blackboard.

Formalism of probability

- I know the formalism for describing chance experiments: outcomes, sample space, events.
- I know the operations of the algebra of events and can use these operations to express complex events in terms of simpler events. For example, the event “at least one of A, B, C occurs” means “ $A \cup B \cup C$ ”.
- I know to what sort of thing we assign a probability.
- I know the rules of probability, including the basic rules and derived rules such as $P(A^c) = 1 - P(A)$ and $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.
- I know the formal definition of “A and B are independent.”
- I know what “A and B are independent” means informally.
- I know when one adds probabilities of events and when one multiplies probabilities of events.

Random variables

- I know the formal definition of a random variable.
- I know the formal definition of the distribution of a random variable.
- I can calculate the distribution of a random variable given a description chance experiment and a random variable for that experiment.
- I can use the “~” notation to indicate that a random variable has one of our named distributions (binomial, Poisson, geometric so far).
- I know the formal definition of the expected value of a random variable.
- I know how to interpret expected value informally, as
- I can calculate the expected value of a random variable given its distribution.
- I know the algebraic properties of expected value.
- I know what the moments of a random variable are, and I can calculate them given the distribution of a random variable using the Law of the Unconscious Statistician (LOTUS).

- I know how to calculate the variance of a random variable X using $E[X^2]$ and $E[X]$.
- I know how to calculate $\sigma(X)$ using $\text{Var}(X)$ for a random variable X .
- I know how to match a random variable described in terms of a chance experiment with the story of the binomial distribution.
- I know the formula for the binomial-distribution probabilities.
- I know the Poisson approximation formula for $\text{binomial}(n, p)$ where n is large and p is small.
- I know the $\text{Poisson}(\lambda)$ distribution.
- I can use the Poisson distributions model counting rare events.
- I know the formula for the $\text{geometric}(p)$ distribution.
- I know how to match the story of a “waiting time” random variable with the $\text{geometric}(p)$ distribution.
- I know the shortcut formulas for expected value and variance of random variables with the $\text{binomial}(n, p)$, $\text{Poisson}(\lambda)$, and $\text{geometric}(p)$ distributions.
- I can solve “urn problems,” matching chance experiments with the story of drawing balls from an urn with a certain number of red balls, using the random variable that counts the number of red balls drawn.

Counting

- I can count using the multiplication rule when counting done in stages with “homogeneity” in the possibility tree at each stage.
- I can count by matching a counting situation with the stories of permutations and combinations.
- I can use Durrett’s terminology and notation for permutations and combinations.
- I can count using the multinomial counting method.
- I can combine counting methods to solve complex problems, such as splitting a class into groups where the order does not matter or making license plates where letters and numbers do not have fixed positions.
- I can use these counting methods to calculate probabilities in situations where all outcomes are equally likely.